

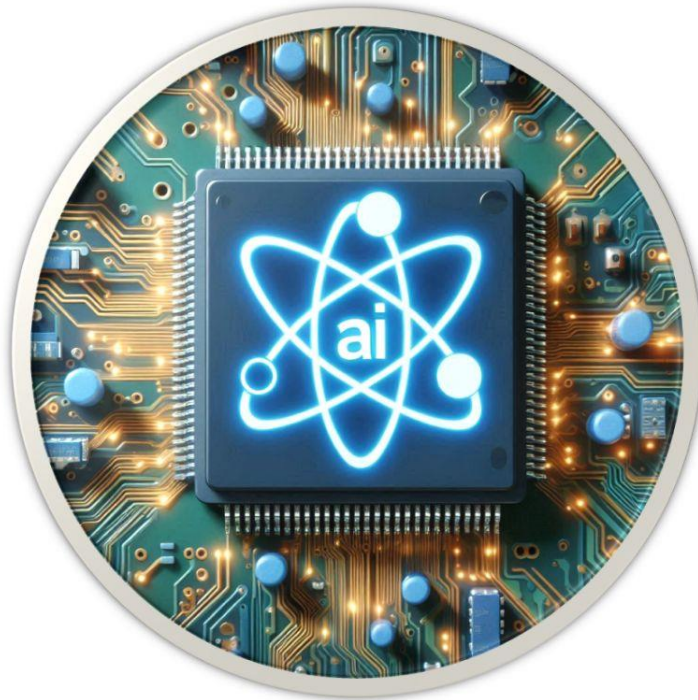
2024

GEOLOGICAL
SURVEY OF INDIA

Hackathon for Innovative
Mineral Hunt Techniques

TEAM AMD

TECHNICAL REPORT



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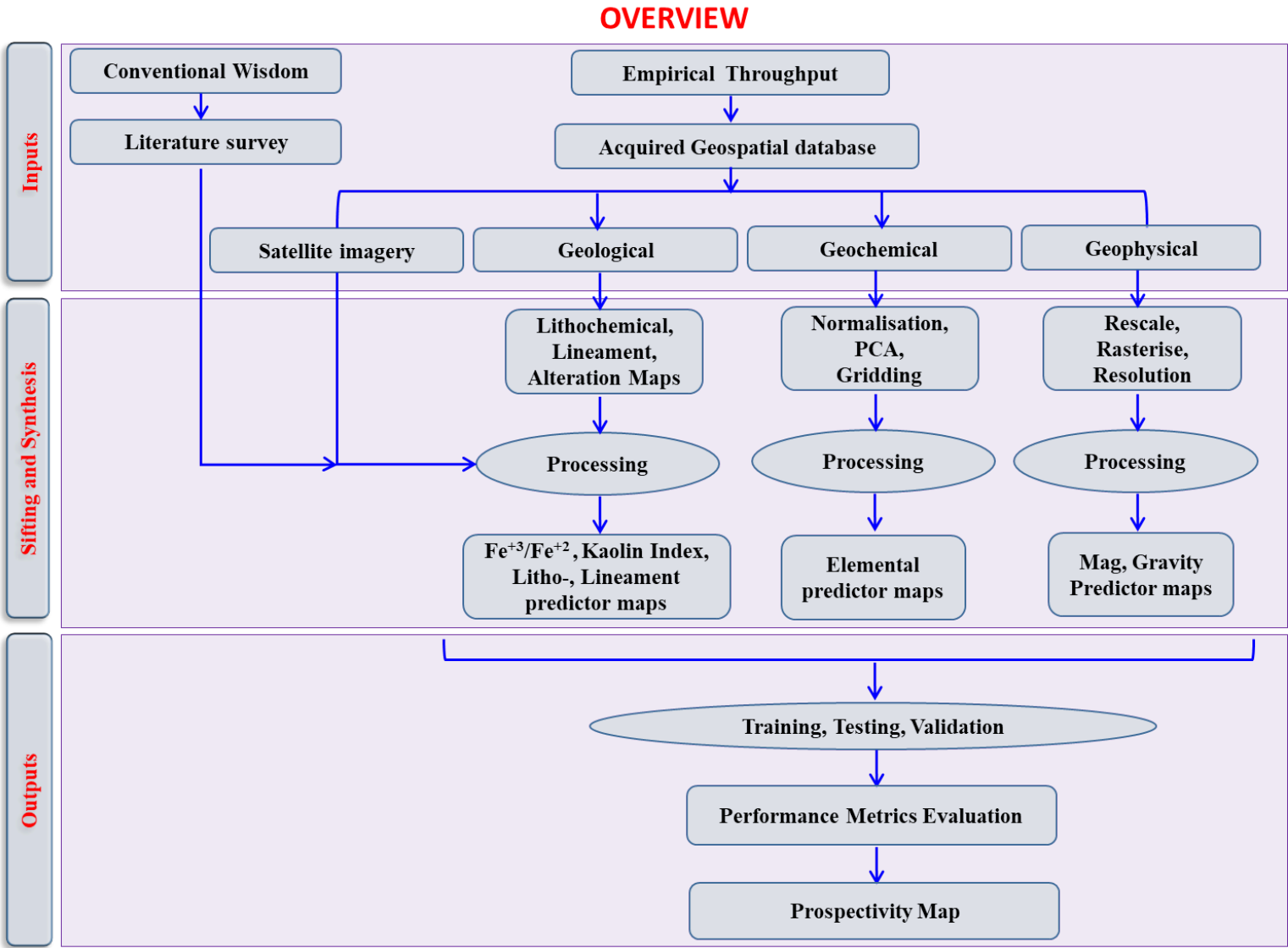
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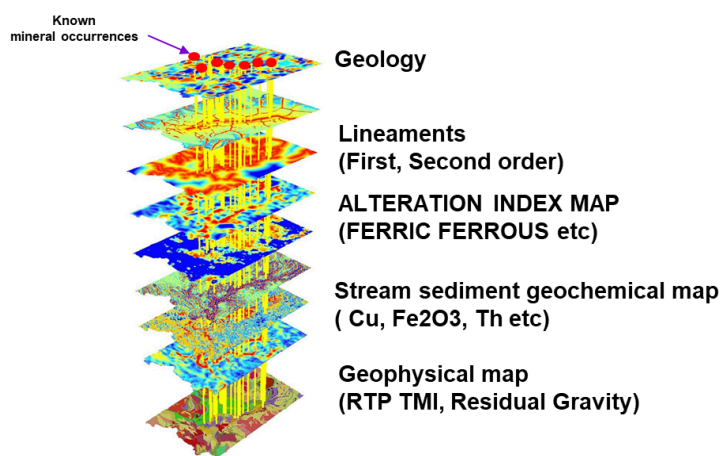
0.OVERVIEW OF ML MODEL



1. Introduction

Problem Statement

The increasing demand for copper and gold necessitates efficient exploration methods that can accurately identify prospective areas for mineral deposits. Traditional prospecting methods often fall short in addressing the complexity and variability of geological factors. This project aims to develop machine learning-based prospectivity mapping algorithms for copper and gold in the Banswara-Bhilwara area of Rajasthan, leveraging bedrock sample data and lithogeochemical information from exploration reports provided by the Geological Survey of India (GSI). By employing advanced machine learning techniques such as Random Forest and XGBoost, the project seeks to enhance the predictive accuracy and efficiency of copper and gold prospectivity mapping. Additionally, Python code will be provided in the form of Jupyter notebooks, utilizing libraries such as `rasterio`, `scikit-learn`, `numpy`, `pandas`, `XGBoost`, and `matplotlib`, with QGIS for spatial analysis and visualization.



Objectives

- To Develop Machine Learning Algorithms:** Create robust algorithms using Random Forest and XGBoost to predict copper and gold prospectivity based on geological and geochemical data.
- To Optimize Predictor Maps:** Identify and select relevant predictor maps based on the mineral systems approach for effective training of the models.
- To Improve Exploration Outcomes:** Increase the accuracy of prospectivity maps, thereby enhancing the efficiency of mineral exploration efforts in the targeted region.

Methodology

1. Data Collection:

- Gather bedrock sample data and lithogeochemical data from exploration reports provided by GSI.
- Compile and preprocess relevant predictor maps based on geological, geochemical, and geophysical features.

2. Data Preparation:

- Clean and preprocess the collected data using Python libraries such as `pandas` and `numpy`.
- Normalize and transform the data as necessary for model training.

3. Model Development:

- Implement machine learning algorithms using `scikit-learn` for Random Forest and `XGBoost` for gradient boosting.
- Train the models on the prepared datasets, adjusting hyperparameters to optimize performance.

4. **Model Evaluation:**

- Evaluate the models using metrics such as accuracy, precision, recall, and F1-score.
- Validate the models against known copper and gold deposit locations.

5. **Map Generation:**

- Generate prospectivity maps using predictions from the trained models.
- Utilize `rasterio` and QGIS for spatial analysis and visualization.

6. **Jupyter Notebooks:**

- Provide Python code in the form of Jupyter notebooks that document the entire process, from data ingestion to model evaluation and map generation, allowing users to replicate and build upon the work.

Deliverables

1. **Machine Learning Models:** Fully developed and trained Random Forest and XGBoost models for copper and gold prospectivity mapping.
2. **Prospectivity Maps:** High-resolution maps indicating areas of varying prospectivity for copper and gold in the Banswara-Bhilwara area.
3. **Technical Report:** A detailed report outlining the methodology, data sources, model performance, and findings.
4. **Jupyter Notebooks:** A collection of Jupyter notebooks containing Python code that illustrates data processing, model training, evaluation, and visualization using the specified libraries.

By implementing this project, we aim to significantly enhance the effectiveness of exploration efforts for copper and gold, facilitating informed decision-making and fostering sustainable mining practices in the region.

2. Building the training location dataset

Generation of dataset of locations is necessary to pick data from feature layers for training. The Training location datasets contains set of points with known mineralization and locations of nonmineralization.

The following details the steps for selections of mineralized and nonmineralized locations and building of dataset.

Methodology

Mineralization locations

All the exploration reports (G2, G3 and G4 stages) were scraped for data regarding locations containing assay data. A cutoff criteria was set to get data of only significant mineralization.

- A total of **265** data points were scraped from the reports associated with **Cu** and **Au - (Cu)** mineralizations. The data points represent collected **Bed rock samples (BSR) and lithogeochemical samples**.
- A final list of points were generated by adding the locations of the mineralization provided in the bhukosh mineralization layer.

The detailed distribution is given below.

S.no	Mineralization	Cutoff	No. of Scraped points	No of Min. locations (with Bhukosh)
1	Cu	0.1wt% or 1000ppm	191	196
2	Au-(Cu)	0.10ppm	83	86

Non mineralized locations

Selection of non mineralized zones is tricky as it is difficult to ascertain with 100% probability that a location is nonmineralized. To tacke the the problem the following was adopted :

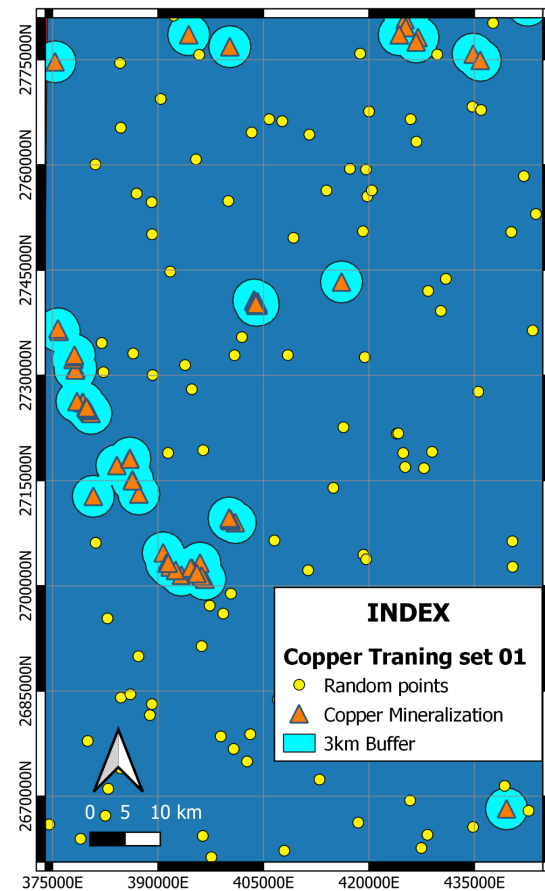
- From the known locations of mineralization a 3km buffer was generated and random points were generated outside those buffer zone.
- Set of Random points generated are equal to the number of points of known mineralization to keep the data set balanced.
- A total of 5 sets of random points were generated for each Cu, Au-(Cu) mineralization.

Final dataset

The below table outlines the building of final datasets of training locations generated by fusing mineralized and nonmineralized locations.

- A total of **5** datasets were generated for each

S.no	Mineralization	Min. locations	Random points	final data set
1	Cu	196	196	392
2	Au-(Cu)	86	86	172



First training set locations of copper containing mineralized and generated random points

3. Feature selection and Feature engineering

Feature selection for copper and gold prospectivity models were carried out using a mineral systems approach. The approach involves identifying key geological, geochemical, and geophysical variables that are indicative of the processes controlling ore deposit formation.

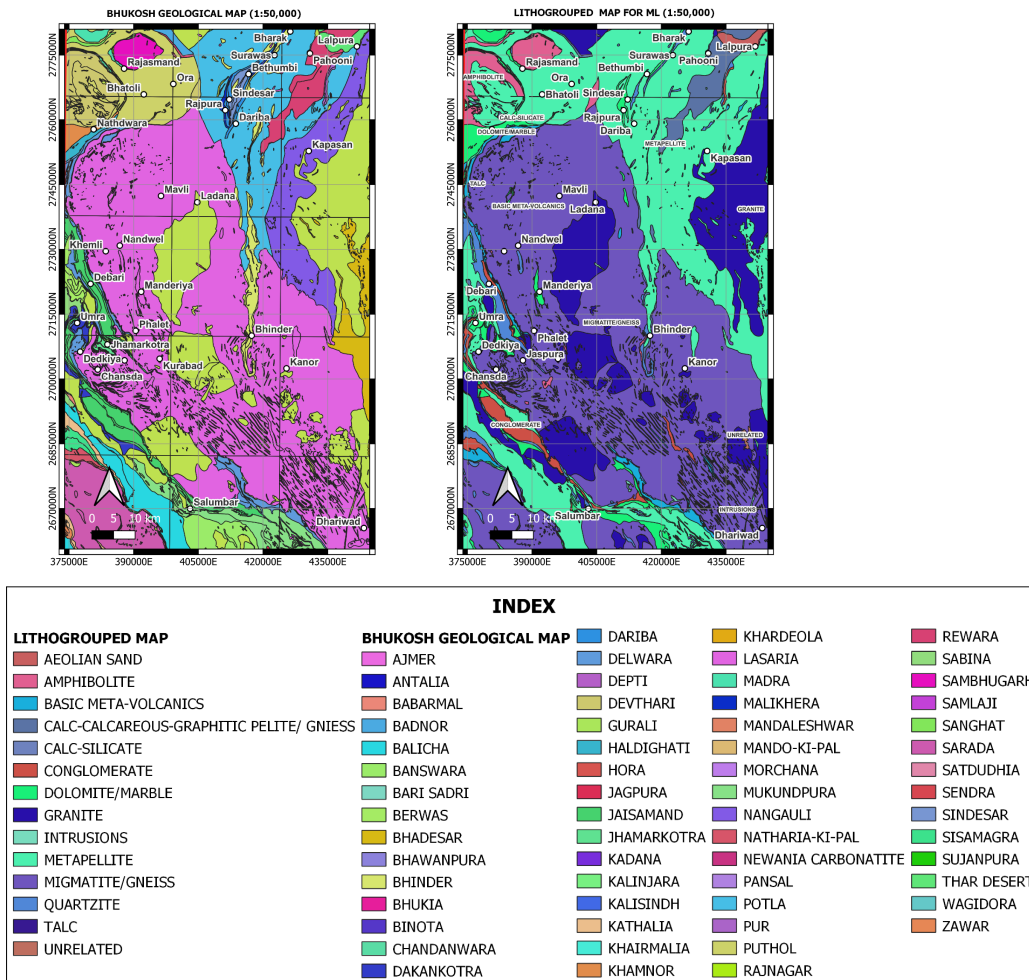
Feature package for model training

In predictive models for copper and gold prospectivity, features are typically grouped based on their geological, structural, geochemical, and geophysical relevance. These feature packages allow the integration of diverse data sources to identify potential mineralization zones. The following table breaks down the typical feature packages used in prospectivity modeling:

Group	Feature Type	Description
Group A: Geology	Lithological units, stratigraphic boundaries, intrusive rocks	Features representing the geological setting that is favorable for copper and gold mineralization, including specific rock types or formations known to host deposits.
Group B: Lineaments (First and Second Order)	Structural controls (faults, shear zones, fractures)	First and second-order lineaments represent major and minor structural features that can control the migration and deposition of mineralizing fluids. First-order features are regional in scale, while second-order features are more localized.
Group C: Alteration Index Maps	Ferric, Ferrous, and Kaolin alteration indices	These maps identify areas where hydrothermal alteration has taken place, which is often associated with ore-forming processes. Ferric (iron oxide), ferrous (iron silicate), and kaolin (clay) alteration are common indicators of copper and gold mineralization.
Group D: Stream Sediment Geochemical Maps	Elemental concentrations (Cu, Au, As, Pb, Zn, etc.)	These maps provide geochemical anomalies derived from stream sediment sampling. They highlight areas with elevated concentrations of metals associated with copper and gold deposits.
Group E: Geophysical Maps	Reduced-to-Pole (RTP) magnetic anomalies, Residual gravity anomalies	Geophysical data such as RTP magnetic maps highlight variations in magnetic properties of the subsurface, while residual gravity anomaly maps reflect density contrasts. Both are useful for identifying subsurface structures, alteration zones, or concealed mineral deposits.

Group A : Geology

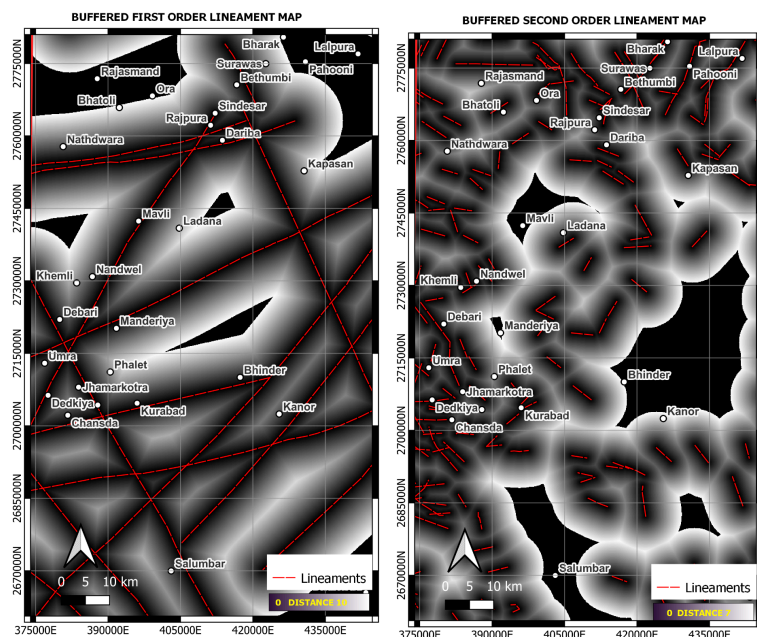
Bhukosh geological map (1:50000) was utilized. The geological map was simplified and grouped based on lithology into **14** lithogroups. The details of grouping of lithologies can seen in the index provided below.



The lithogrouped map was rasterized, rescaled and resampled to 100mx100m before inserting into the model

Group B : Lineaments

Bhukosh lineaments map was utilized (1:50,000 scale). Lineaments based on the proximity and size were classified into first or second order lineaments. The lineaments from the map were buffered 10km for first order lineaments and 7km for second order lineaments.

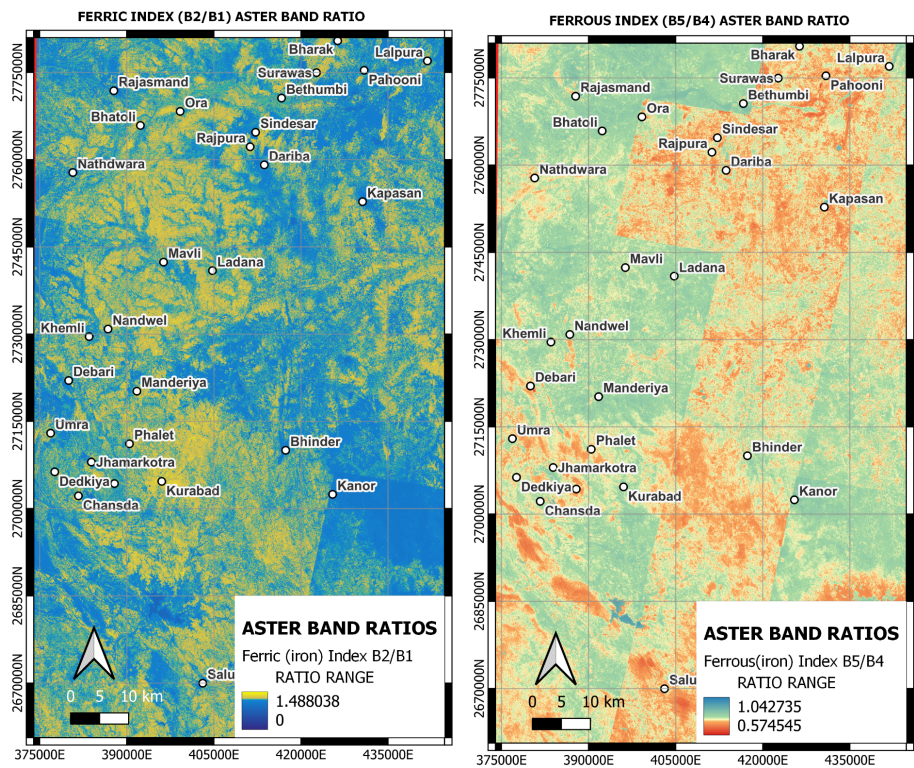


Lineament map was rasterized, rescaled and resampled to 100x100 before inserting into model

Group C : Alteration index maps

Alteration index maps with special emphasis on iron was utilized as predictive maps. ASTER band ratios were chosen for building alteration indices maps. Following table shows the predictor maps utilized

S.no	Index	Ratio
1	Ferric	Band 2 / Band 1
2	Ferric	Band 4 / Band 3
3	Ferrous	Band 5 / Band 4
4	Kaolin	Band 6 /Band 5

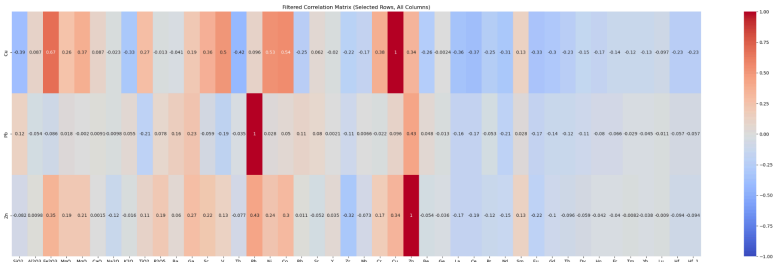


Group D : Stream sediment geochemical maps

Copper and Gold models selection of geochemical elements as predictor variable were done based on Correlation matrix and Principal component analysis (PCA). Major element data was standardized. Trace element data were lognormalized before standardizing. Efforts were made to make the distributions as normal as possible.

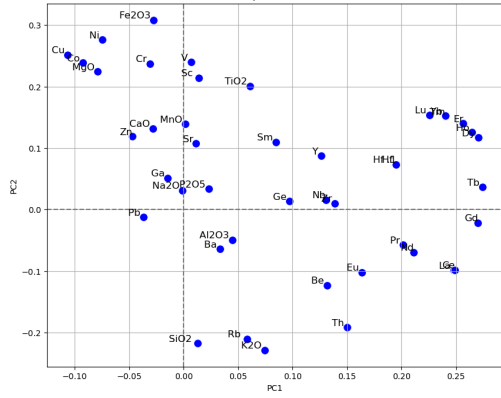
Model	Elements utilized
Copper	Co, CrO, Cu, Fe2O3, MgO ,V , Zn
Gold	CaO, Cr, Cu, Fe2O3, MgO, Na2O, Th, TiO2

All the elements were gridded, rasterized, rescaled and resampled prior to application of model

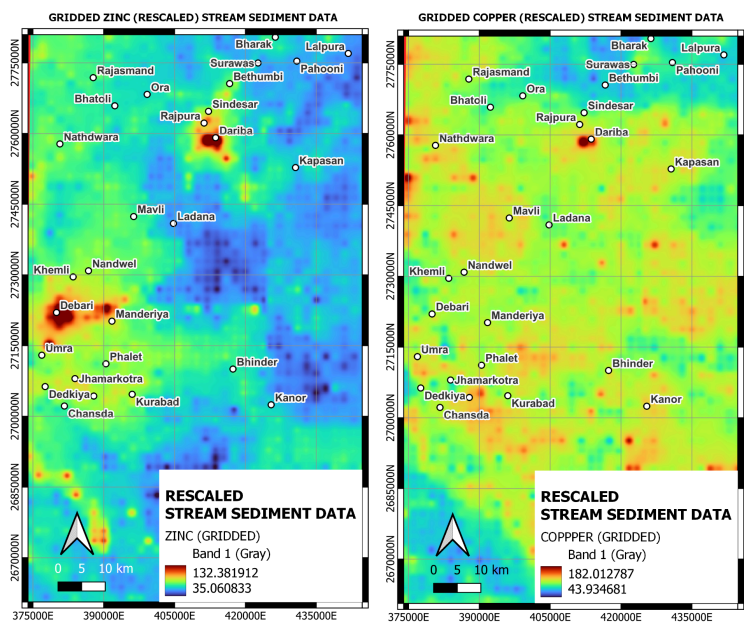


Correlation matrix for Copper

PCA Biplot of Features

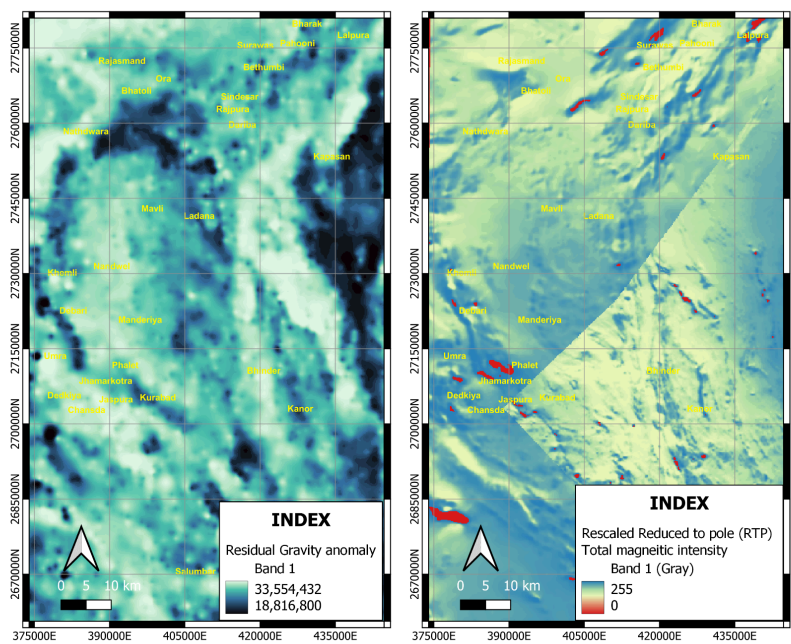


PCA for Copper



Group E : Geophysical maps

Aeromagnetic and gravity maps were processed. Aeromagnetic maps were reduced to pole them merged, rescaled and resampled. Gravity maps were processed to build residual gravity maps upto 500m.

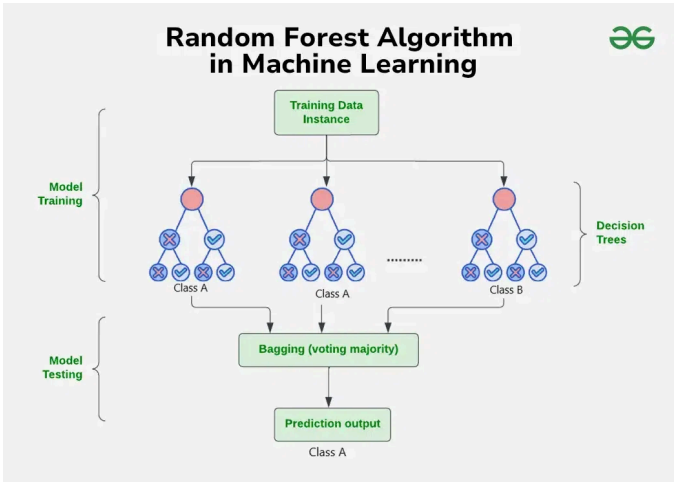


4. Model selection

For our copper and gold prospectivity mapping, we have chosen to utilize two powerful ensemble decision tree methods: **Random Forest (RF)** and **XGBoost**. These algorithms have been selected for their ability to handle complex, high-dimensional data and for their proven effectiveness in producing accurate, robust predictions. Below, we explain how each algorithm works and why they are critical for our prospectivity modeling efforts.

Random Forest (RF) Algorithm

The **Random Forest** algorithm is a versatile and widely used machine learning technique for classification and regression tasks. It builds multiple decision trees and aggregates their outputs to generate a more reliable prediction. Here's how it operates:



- **Tree Construction:** During model training, Random Forest constructs several decision trees using random subsets of the data. Each tree is trained on a bootstrap sample, which is a randomly selected portion of the training data. This process is known as "bagging" (bootstrap aggregating).
- **Random Feature Selection:** At each node of the tree, a random subset of features (such as geological, geochemical, or geophysical variables) is considered for splitting the data. This randomness ensures that the trees are not correlated and reduces overfitting.
- **Prediction:** Once all trees are built, their individual predictions are combined. For classification (e.g., predicting whether a location is prospective for copper or gold), Random Forest takes the majority vote across all trees. For regression (e.g., predicting the likelihood of mineralization), it averages the outputs of all trees.
- **Feature Importance:** A valuable byproduct of the Random Forest algorithm is the importance ranking of the input features. This helps us identify which variables, such as specific alteration indices or geophysical anomalies, are the most significant in predicting mineralization zones.

Importance in Prospectivity Mapping:

- **Handling High-Dimensional Data:** Random Forest can efficiently handle large datasets with many features, which is common in mineral exploration when integrating geological, geochemical, and geophysical layers.
- **Resistance to Overfitting:** The inherent randomness in tree construction and feature selection ensures that the model generalizes well, even when dealing with noisy or incomplete exploration data.
- **Interpretability:** The algorithm's ability to rank features provides insights into the most critical factors controlling mineralization, which is valuable for targeting exploration efforts.

XGBoost Algorithm

XGBoost (Extreme Gradient Boosting) is an advanced machine learning algorithm known for its speed and accuracy. It enhances traditional gradient boosting methods by incorporating several optimizations, making it highly effective for complex and large datasets. Here's how XGBoost works:



- **Boosting Approach:** Unlike Random Forest, where trees are built independently, XGBoost builds decision trees sequentially. Each new tree focuses on correcting the errors made by the previous ones. In this way, XGBoost "boosts" the performance of the model by continuously refining it.
- **Gradient Descent Optimization:** XGBoost uses gradient descent to minimize the errors of the model by adjusting tree weights. It continuously updates the model by following the gradient of a loss function, which measures how well the predictions match the actual outcomes.
- **Regularization:** To prevent overfitting, XGBoost incorporates regularization techniques (L1 and L2 regularization). These techniques control the complexity of the model, especially when dealing with noisy datasets or when many features are correlated.
- **Handling Missing Data:** XGBoost has built-in mechanisms to handle missing values, a common issue in geological and exploration data. This capability ensures that the model can still make accurate predictions even when parts of the dataset are incomplete.

Importance in Prospectivity Mapping:

- **High Precision and Efficiency:** XGBoost is known for its superior performance on large and complex datasets, making it ideal for mineral prospectivity mapping, where multiple data layers need to be integrated and processed quickly.
- **Adaptability to Complex Relationships:** The algorithm is well-suited for capturing complex, non-linear relationships between geological features and mineralization, which is crucial for identifying prospective copper and gold targets.
- **Robustness Against Overfitting:** XGBoost's regularization techniques help ensure that the model is not overly tailored to training data, allowing it to perform well on unseen exploration areas.

Why We Chose These Methods

In the field of mineral prospectivity mapping, where we work with large, complex, and multi-dimensional datasets (geology, lineaments, geochemistry, geophysics), both Random Forest and XGBoost offer unique advantages:

- **Random Forest** provides excellent interpretability through feature importance rankings and is robust against overfitting.

- **XGBoost** is highly efficient, precise, and adaptive to complex patterns in the data, making it ideal for handling the intricate relationships between geological and mineralization processes.

By utilizing both methods, we can ensure that our prospectivity model captures all critical factors controlling copper and gold mineralization, while producing reliable, high-quality predictions for exploration targeting.

5. Copper Prospectivity model (CPM)

Results of Copper Prospectivity Model (CPM)

In the development of the Copper Prospectivity Model (CPM), we employed two advanced ensemble decision tree algorithms: **Random Forest** and **XGBoost**. Both models were trained on 5 training sets (TS01 to TS05) datasets over 17 predictor maps of geological, geochemical, and geophysical features relevant to copper mineralization.

Random Forest Model

- **Five Training Sets:** We utilized five different training sets to construct individual Random Forest models, allowing us to assess the stability of predictions across various data splits.
- **Testing and Validation Sets:**
 - We employed an **80:20** split for the training and testing sets, ensuring that 20% of the data was reserved for validation. This 20% was further divided into two separate validation sets, providing a robust assessment of the models on unseen data.
 - This structure ensures that our models were evaluated on data they have not encountered during training, enhancing the reliability of the predictive capabilities.
- **Average Accuracy:** The average accuracy of the Random Forest model across the five training sets was **91.8%**, indicating a strong capability in correctly predicting copper mineralization.
- **Precision and Recall:** The model demonstrated high precision and recall rates, reflecting a low false-positive rate and effective coverage of actual prospective zones.

Performance Metrics for Random Forest

The following table summarizes the performance metrics for the Random Forest model across the five training sets, including accuracy and ROC AUC scores for two validation sets:

Performance Metrics for Training Sets

1. **High Accuracy:** An average accuracy of **91.8%** indicates the model effectively classifies areas for copper prospectivity, minimizing misclassifications.
2. **Excellent Discrimination:** The average ROC AUC score of **95.1%** demonstrates strong capability in distinguishing between high and low prospectivity areas, reinforcing the model's reliability.
3. **Consistent Robustness:** Despite slight variability across training sets, the model consistently performs well, showcasing its effectiveness in handling different data distributions and geological characteristics.

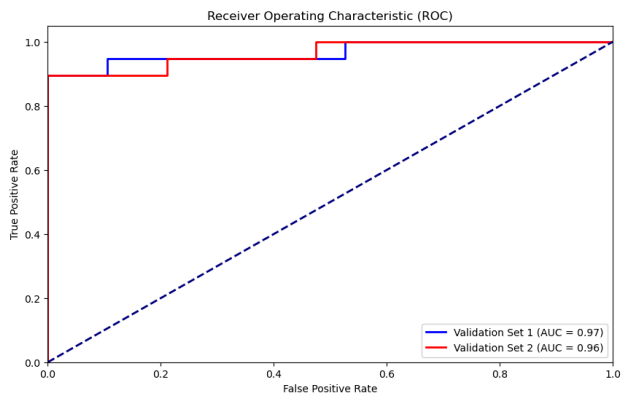
Training Set	Validation Set 1	Validation Set 2
TS01		
Accuracy Score	89	89
ROC AUC Score	95	94
TS02		
Accuracy Score	95	92

Training Set	Validation Set 1	Validation Set 2
ROC AUC Score	96	96
TS03		
Accuracy Score	87	93
ROC AUC Score	94	96
TS04		
Accuracy Score	92	92
ROC AUC Score	95	95
TS05		
Accuracy Score	87	90
ROC AUC Score	95	96
Average	91.8	95.1

Classification Report for Random Forest (TS02)

Class	Precision	Recall	F1-Score	Support
0	0.90	0.95	0.92	19
1	0.94	0.89	0.92	19
Accuracy			0.92	38
Macro Avg	0.92	0.92	0.92	38
Weighted Avg	0.92	0.92	0.92	38

Receiver operating Characteristic (ROC) for Random Forest (TS02)



Stacked Results for Random Forest

To synthesize the results from the five training sets, we implemented a **stacking approach** where the outputs of each model were layered to generate cumulative scores. Each location was assigned a score from **0 to 5**:

- **Score 0**: No models predicted the area as prospective (very low potential).
- **Score 5**: All five models predicted the area as prospective (very high potential).

This approach allowed for the creation of a detailed prospectivity map categorized into five classes based on the likelihood of hosting copper deposits.

XGBoost Model

The **XGBoost** algorithm was also employed to enhance the predictive capabilities of the CPM. Here's how it was implemented:

- **Training Process:** Similar to the Random Forest model, we trained the XGBoost model on the same five training sets. Each model was built sequentially, focusing on correcting the errors of the previous iterations to improve prediction accuracy.
- **Hyperparameter Tuning:** We utilized Grid Search Cross-Validation (CV) to systematically optimize hyperparameters such as learning rate, maximum depth, and regularization. This tuning process significantly improved the model's performance and generalization.
- **Performance Metrics:** The XGBoost model achieved comparable performance, with an average accuracy of approximately **89.4%** across the training sets. Its precision and recall rates further reinforced the model's reliability in identifying prospective areas for copper mineralization.

Interpretation of Combined Models

- **Very High Potential Zones:** Areas scoring a 5 in either model are deemed highly prospective for copper exploration.
- **Moderate to Low Potential Zones:** Scores ranging from 1 to 3 suggest some mineralization potential but with lower confidence levels.

Performance Metrics for Training Sets

- **High Accuracy: 89.4%**—indicates high reliability in predicting copper mineralization.
- **High ROC AUC: 95.0%**—demonstrates excellent ability to distinguish between high and low prospectivity areas.
- **Strong Consistency:** Accuracy scores ranged from 86% to 92%, showcasing robust performance across different training sets.
- **High Discriminatory Power:** ROC AUC scores ranged from 93% to 97%, reinforcing the model's precision in prospectivity mapping.

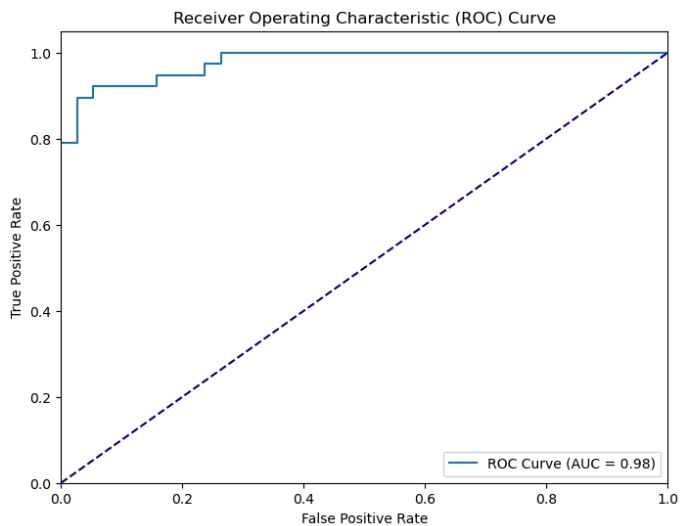
Training Set	TS01	TS02	TS03	TS04	TS05	Average
Accuracy Score	86	92	88	92	89	89.4
ROC AUC Score	93	97	94	96	95	95.0

Classification report for XGBoost (TS02)

Metric	Precision	Recall	F1-Score	Support
Class 0	0.88	0.97	0.93	38
Class 1	0.97	0.87	0.92	38
Accuracy			0.92	76

Metric	Precision	Recall	F1-Score	Support
Macro Avg	0.93	0.92	0.92	76
Weighted Avg	0.93	0.92	0.92	76

Reciver Operating Characteristic (ROC) Curve for XGBoost (TS02)



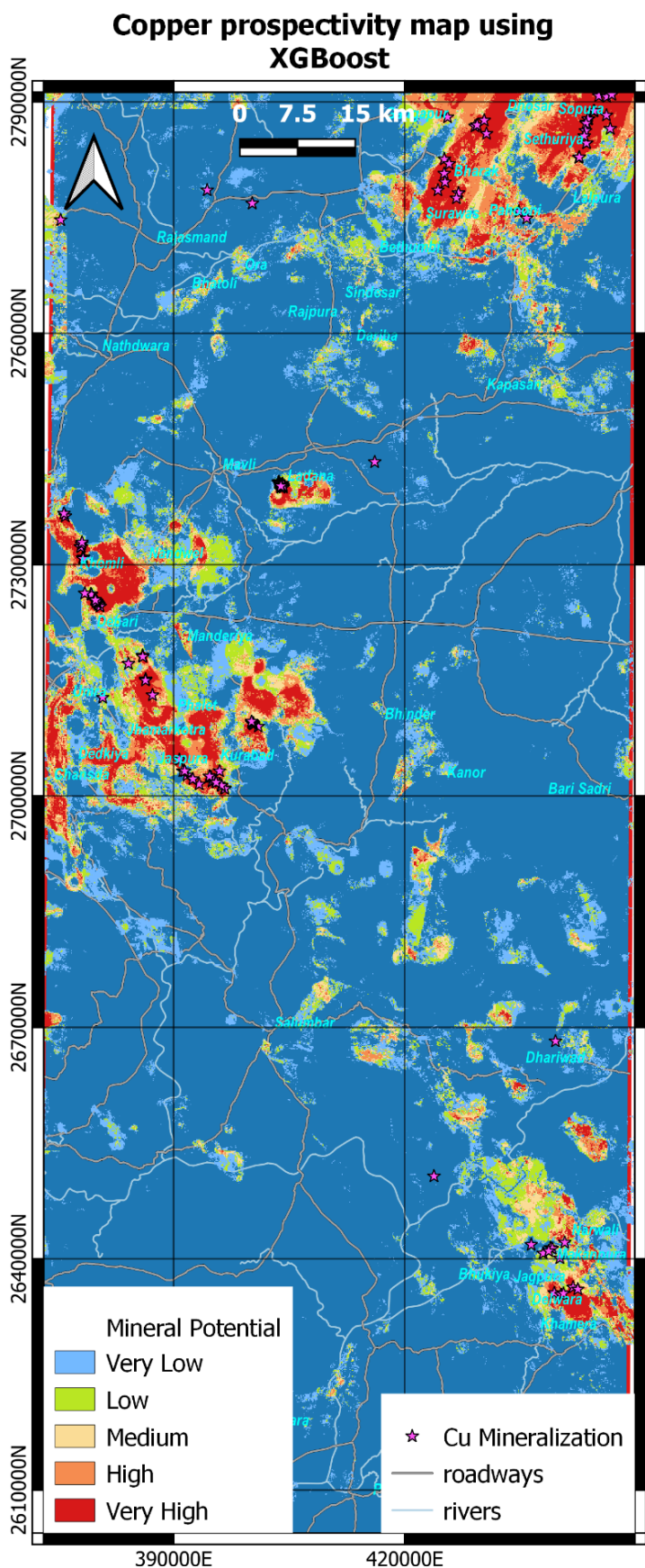
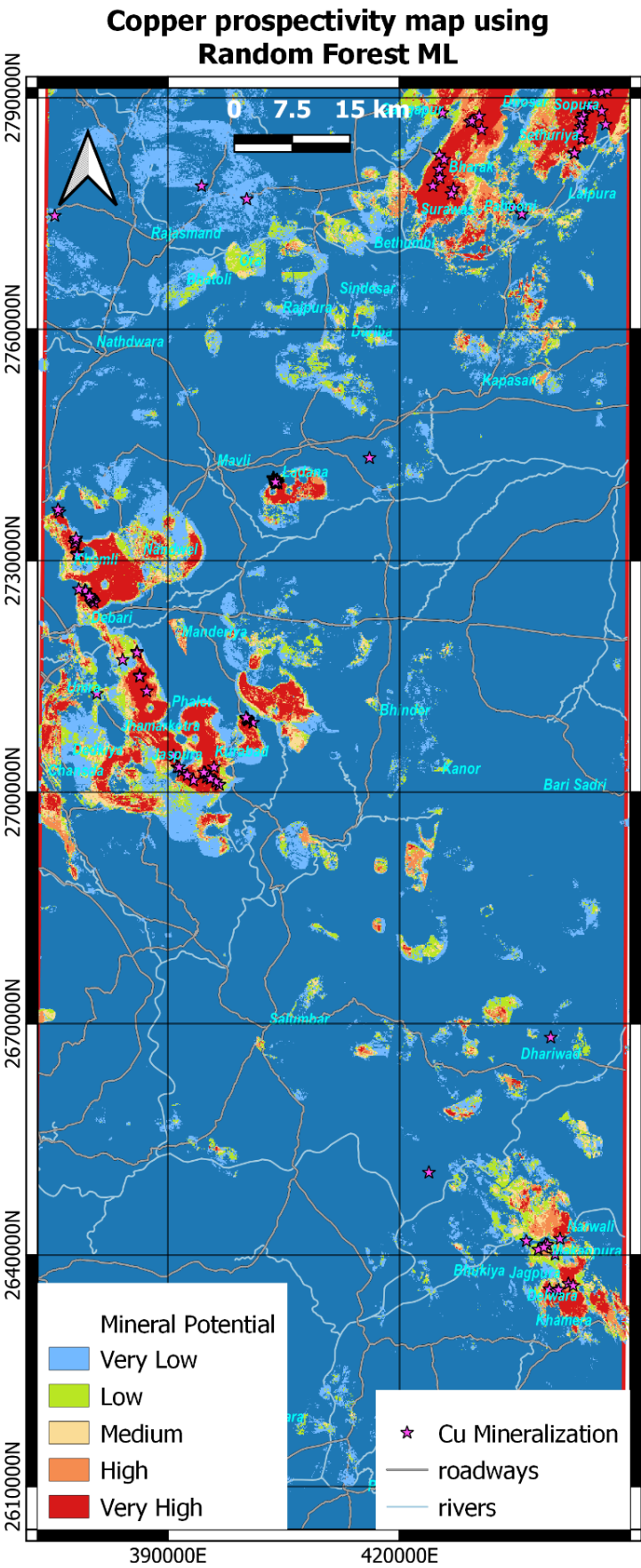
Stacked Results for XGBoost

Just like with Random Forest, we utilized a stacking approach for the XGBoost model, where the outputs from the five models were combined to produce cumulative scores:

- **Score 0:** No models predicted the area as prospective (very low potential).
- **Score 5:** All five models predicted the area as prospective (very high potential).

This stacking strategy enabled us to develop a comprehensive prospectivity map that integrated the predictions from both Random Forest and XGBoost

5a. Copper prospectivity maps



6. Gold Prospectivity model (GPM)

The development of the Gold Prospectivity Model (GPM) utilized two advanced ensemble decision tree algorithms: **Random Forest** and **XGBoost**. Both models were trained on five distinct training sets (TS01 to TS05), incorporating 17 predictor maps covering geological, geochemical, and geophysical features crucial for identifying gold mineralization.

Random Forest Model

- **Five Training Sets:** Each of the five training sets was used to construct individual Random Forest models, enabling us to assess the stability of predictions across varying data distributions.
- **Testing and Validation Sets:**
 - An **80:20** data split was employed for training and testing, with 20% reserved for validation. This subset was divided into two validation sets to enhance the robustness of the model evaluations.
 - This strategy ensured that model performance was evaluated on unseen data, increasing confidence in its predictive capabilities.
- **Average Accuracy:** The Random Forest model achieved an impressive **93.8%** accuracy across the five training sets, demonstrating strong predictive power for gold mineralization.
- **Precision and Recall:** The model exhibited high precision and recall, indicating a low false-positive rate and effectively identifying prospective gold zones.

Performance Metrics for Random Forest

The table below summarizes the performance of the Random Forest model across the five training sets, providing accuracy and ROC AUC scores for both validation sets:

- **Overall Average Accuracy:** 93.8%
- **Overall Average ROC AUC Score:** 97.8%

Training Set	Val.Set1 (Accuracy)	Val.Set2 (Accuracy)	Val.Set1 (ROC AUC)	Val.Set2 (ROC AUC)
TS01	96	96	99	98
TS02	92	100	95	100
TS03	92	92	98	98
TS04	88	100	96	100
TS05	94	88	97	97
Average	92.0	95.2	97.0	98.6

Classification Report for Random Forest (TS02 Val Set 1)

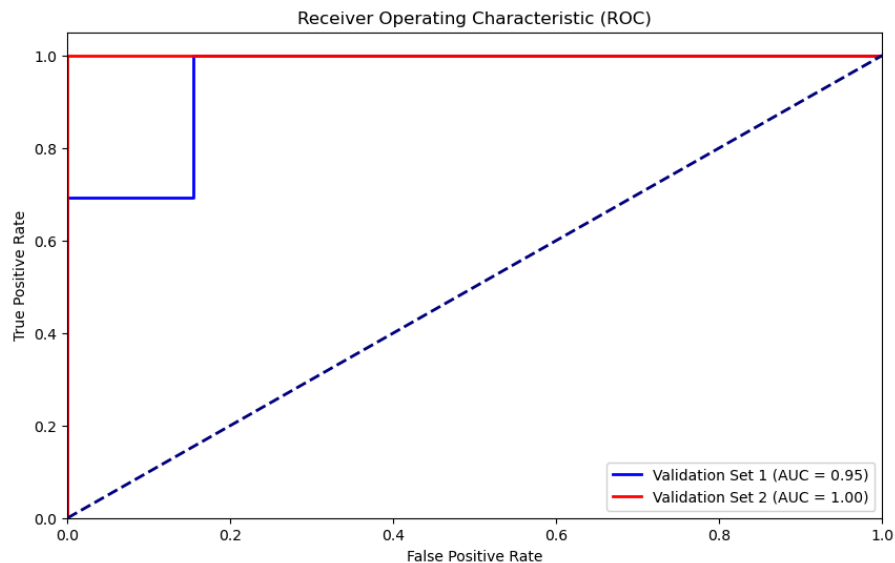
Class	Precision	Recall	F1-Score	Support
0	1.00	0.85	0.92	13

Class	Precision	Recall	F1-Score	Support
1	0.87	1.00	0.93	13
Accuracy			0.92	26
Macro Avg	0.93	0.92	0.92	26
Weighted Avg	0.93	0.92	0.92	26

- **Accuracy Score: 0.9231**
- **ROC AUC Score: 0.9527**

The table reveals consistently strong performance, with high accuracy and excellent ROC AUC scores across different validation sets, indicating the model's strong discriminatory power in identifying high and low prospectivity areas.

Receiver Operating Characteristic (ROC) for XGBoost (TS02)



Stacked Results for Random Forest

To synthesize the predictions from the five training sets, a **stacking approach** was used. This method layered the model outputs to generate cumulative scores, where each location was assigned a score from **0 to 5**:

- **Score 0:** No models identified the area as prospective (very low potential).
- **Score 5:** All five models predicted the area as prospective (very high potential).

This approach enabled the creation of a detailed prospectivity map, categorizing locations based on their likelihood to host gold deposits.

XGBoost Model

The **XGBoost** algorithm was utilized to enhance the predictive capabilities of the Gold Prospectivity Model (GPM). Below is an overview of its implementation and performance metrics.

Implementation Details

1. Training Process:

- The XGBoost model was trained on the same five training sets as the Random Forest model.

- The training process involved iteratively refining predictions by correcting errors from previous iterations.

2. Hyperparameter Tuning:

- **Grid Search Cross-Validation (CV)** was employed to optimize hyperparameters such as learning rate, maximum depth, and regularization.
- This tuning process significantly enhanced the model’s performance and generalization ability.

Performance Metrics

The XGBoost model demonstrated robust performance in predicting gold prospectivity:

- **Average Accuracy: 90.6%**
- **Average ROC AUC Score: 95.4%**

Training Set	TS01	TS02	TS03	TS04	TS05	Average
Accuracy Score	91	91	86	94	91	90.6
ROC AUC Score	96	95	89	98	99	95.4

Classification Report for XGBoost (TS04)

Here is the classification report for the XGBoost model on the TS04 validation set:

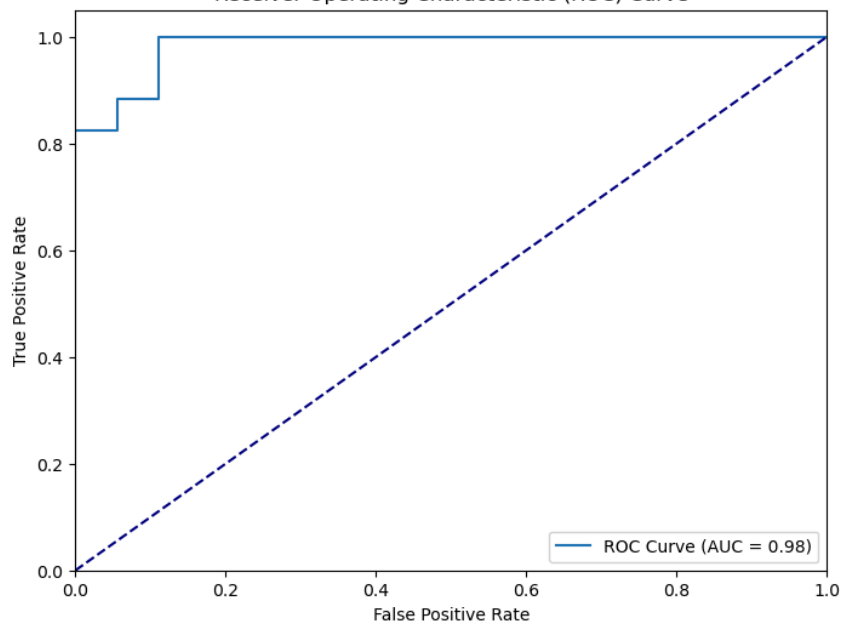
Metric	Precision	Recall	F1-Score	Support
Class 0	1.00	0.89	0.94	18
Class 1	0.89	1.00	0.94	17
Accuracy			0.94	35
Macro Avg	0.95	0.94	0.94	35
Weighted Avg	0.95	0.94	0.94	35

Performance Metrics

- **Accuracy Score: 94.29%**
- **ROC AUC Score: 98.37%**

Receiver Operating Characteristic (ROC) Curve for XGBoost (TS04)

Receiver Operating Characteristic (ROC) Curve



Stacked Results for XGBoost

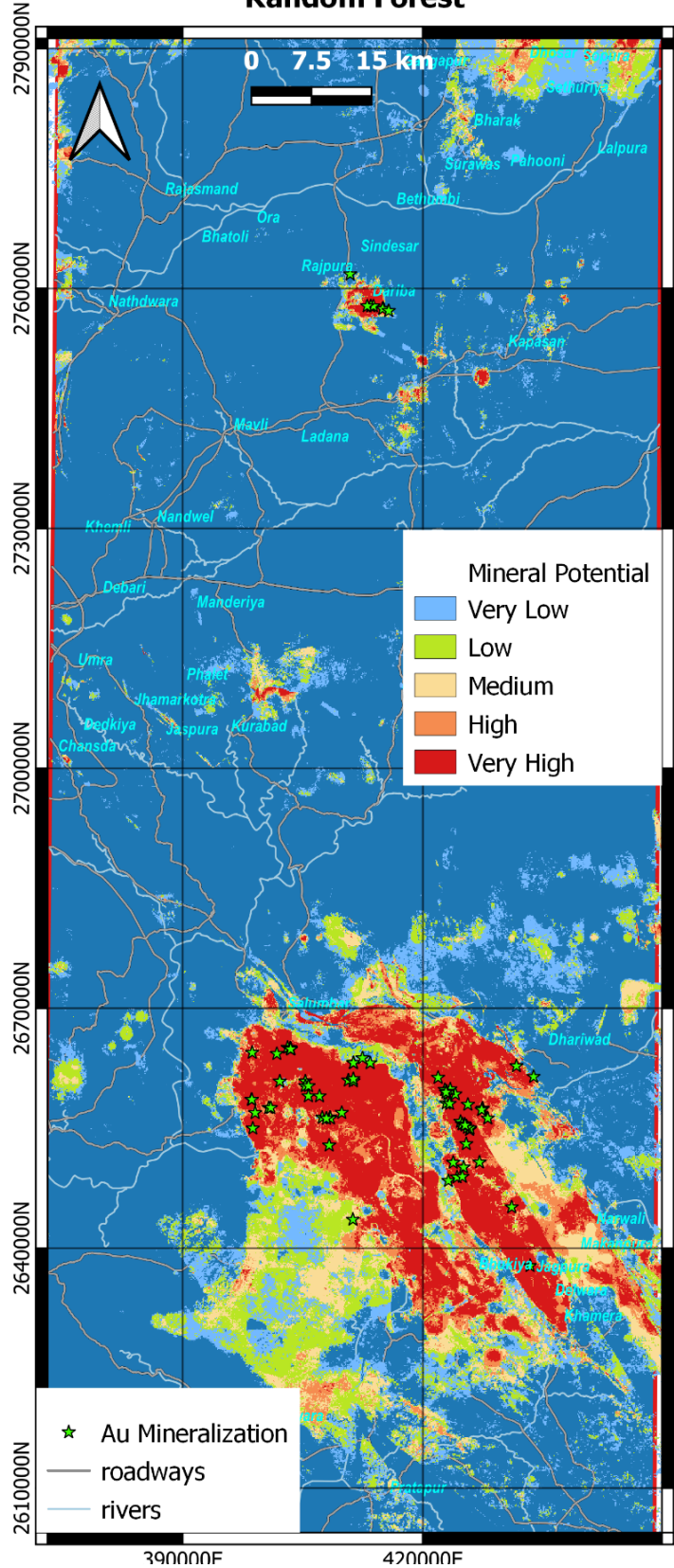
The stacking approach was also applied to the XGBoost model, combining the outputs from the five models to generate cumulative scores:

- **Score 0:** No models predicted the area as prospective (very low potential).
- **Score 5:** All models predicted the area as prospective (very high potential).

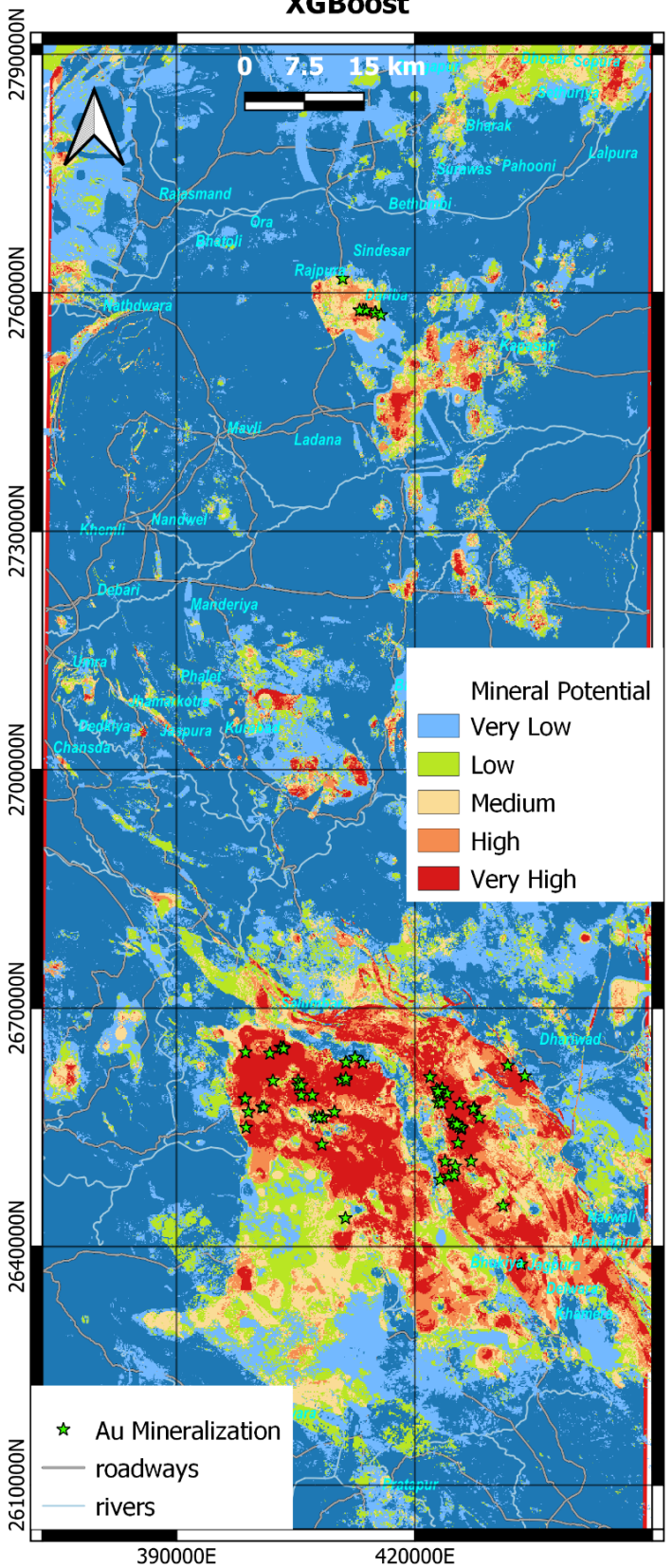
This comprehensive evaluation highlights the effectiveness of the XGBoost model in identifying gold prospectivity and confirms its reliability for exploration purposes.

6a. Gold Prospectivity maps

Gold prospectivity map using Random Forest



Gold prospectivity map using XGBoost



7. Interpretation

Interpretation of Copper and Gold Prospectivity Maps

In mineral exploration, prospectivity maps help identify areas with high potential for mineral deposits based on geological, geochemical, and geophysical data. In this case, the copper and gold prospectivity maps were derived using two machine learning models: **XGBoost** and **Random Forest**, with both models outputting a score from 0 to 5, where 5 represents the highest prospectivity.

Method of Stacking Maps

The approach used to combine the results from both models is a **stacking** method. The stacked maps represent areas where both models agree on high mineral potential. Here's how the method works:

- Each model's output is scaled from **0 to 5**, with **5 representing high potential** and **0 representing low potential**.
- Only areas where **both the XGBoost map and the Random Forest map score a 5** are selected for further exploration.

Interpretation:

Areas with High Agreement (5 on both models):

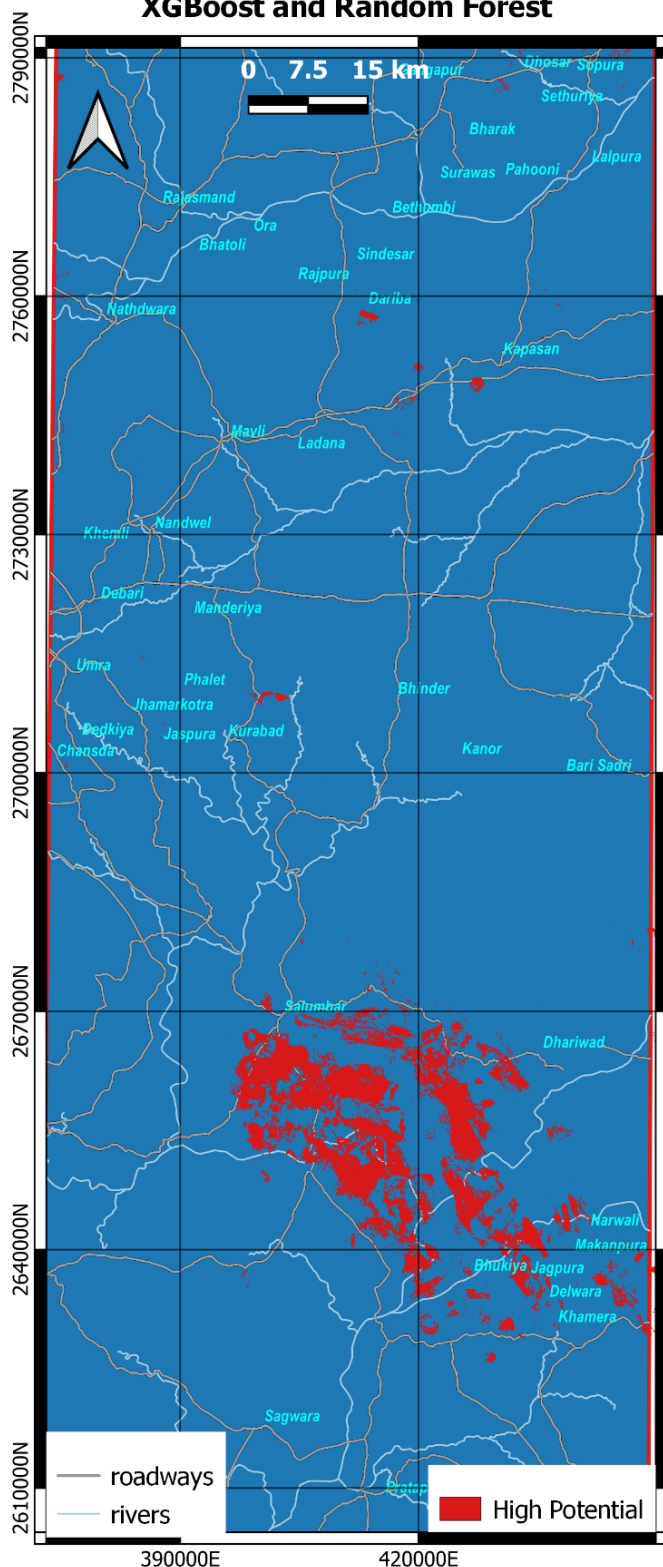
- - These areas are considered to have the highest potential for hosting copper and/or gold deposits.
 - Both the machine learning models independently predict high mineral prospectivity, increasing confidence in the potential of these zones.
 - These high-priority targets should be the focus of detailed exploration, including geological mapping, geochemical sampling, and geophysical surveys.

Proposed Target Areas for Gold Exploration

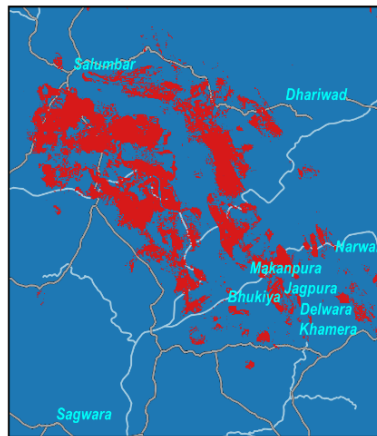
Based on the combined XGBoost and Random Forest prospectivity maps, the following high-potential areas for **gold** exploration are identified:

1. **Salumbar-Dhariwad-Barwali Extension of Bhukiya-Japura**
2. **Dhosar-Sopura Block**
3. **Phalet-Kurabad Block**
4. **Kapasan-South Dariba Block**

Cumulative Gold prospectivity map of XGBoost and Random Forest



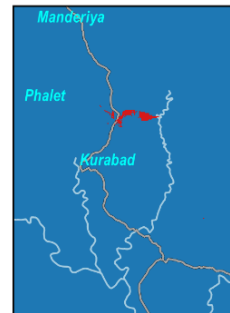
Salumbar-Dhariwad-Narwali extension



Dhosar-Sopura block



Phalet-Kurabad block



Kapasan-South Dariba block



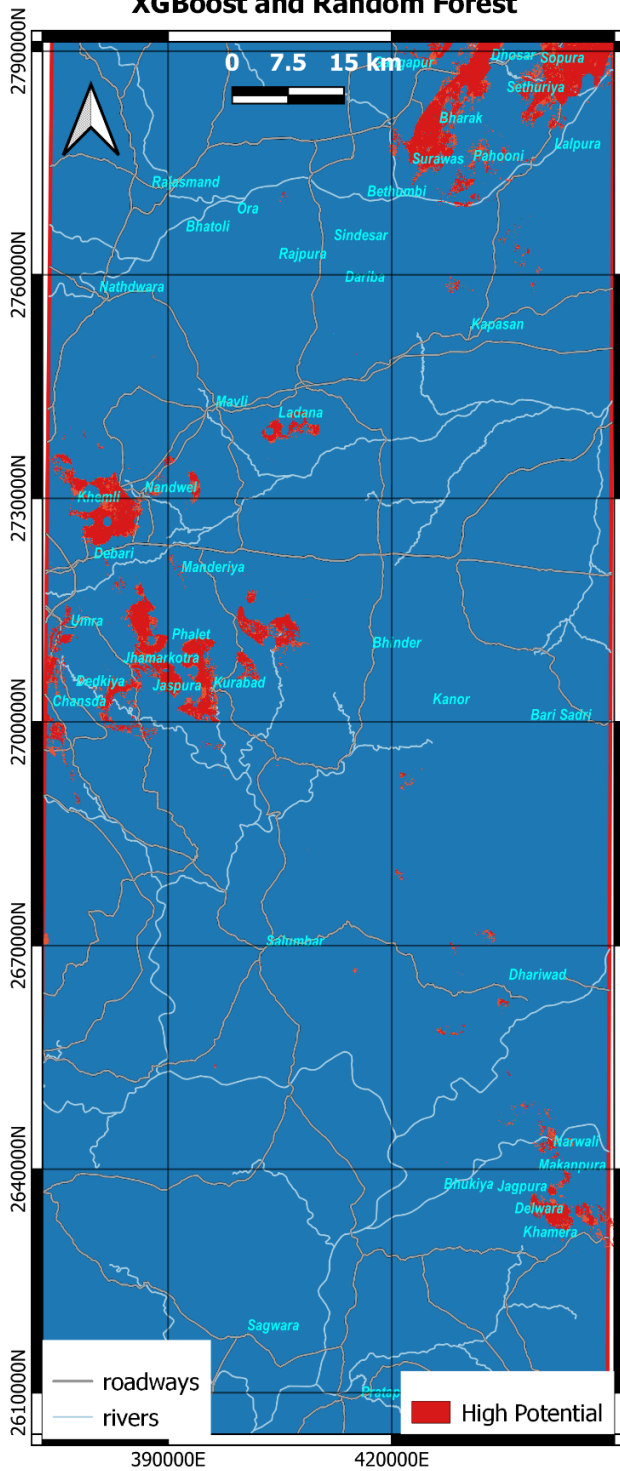
Proposed Target Areas for Copper Exploration

For **copper** exploration, the following high-prospectivity areas are prioritized:

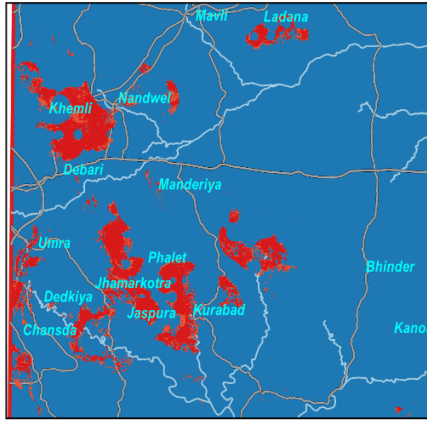
1. Khemli-Nandwel-Kurabad Extension
2. Narwali Block and Dhariwad Block
3. Gangapur-Surawas-Pahooni Block

These areas, showing high agreement (score 5 on both models), should be prioritized for detailed mapping, geochemical sampling, and geophysical surveys.

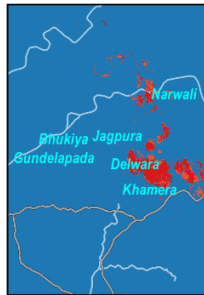
Cumulative Copper prospectivity map of XGBoost and Random Forest



Khemli - Nandwel - Kurabad extension



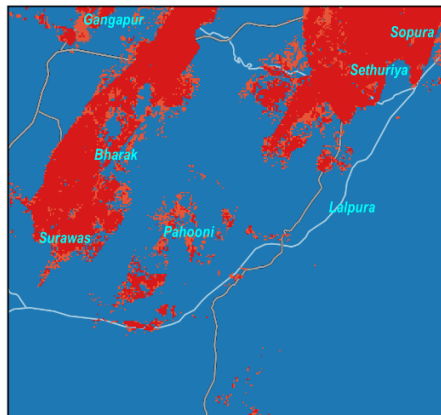
Narwali block



Dhariwad block



Gangapur-Surawas-Pahooni block



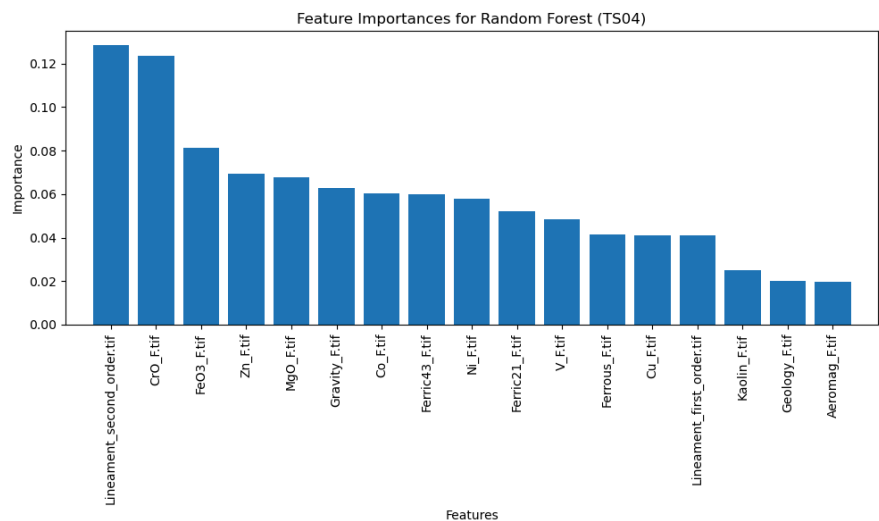
8. Feature Importance Study

Feature importance study

Feature importance studies are critical in mineral exploration as they help identify and prioritize geological, geochemical, and geophysical parameters that influence the presence of mineral deposits. By utilizing machine learning models such as Random Forest and XGBoost, these studies enable geologists to assess various features that contribute to mineral prospectivity effectively.

Feature Importance Analysis for Copper Prospectivity (Random Forest & XGBoost)

Random Forest Key Features:



- **Second Order Lineaments:**

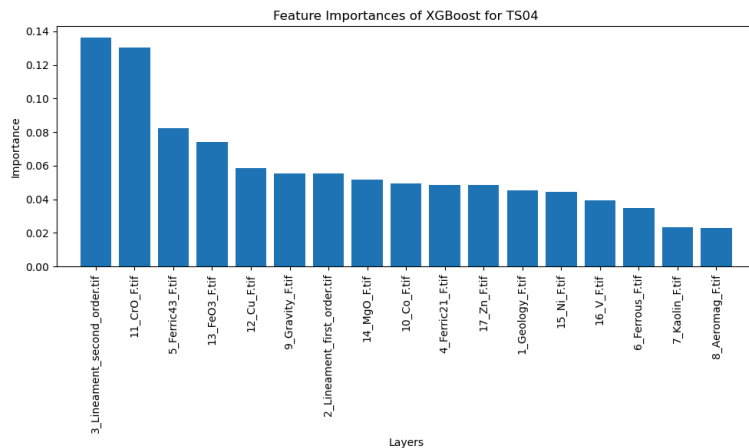
- These structural features, including minor faults and fractures, are crucial for the movement of mineralizing fluids. Their high importance underscores their role in localizing copper deposits, as they often provide pathways for hydrothermal activity.
- **Copper (Stream Sediment):**
 - The concentration of copper in stream sediments serves as a direct geochemical indicator of nearby copper mineralization. High levels of copper in sediments can highlight areas of potential interest for further exploration and indicate active mineralization processes.
- **Fe₂O₃ (Iron Oxide):**
 - The presence of iron oxides is a strong proxy for hydrothermal alteration zones. These zones are often associated with copper mineralization, and thus Fe₂O₃ serves as an important marker for identifying potential ore-bearing areas.
- **Cr₂O₃ (Chromium Oxide):**
 - The presence of chromium oxide can indicate the potential for specific types of mineralization, as it is often associated with the formation of certain ore bodies. Its signal in geochemical analyses can provide additional information about mineralizing processes and associated minerals.
- **Residual Gravity:**

- Variations in residual gravity measurements can reveal subsurface density anomalies that are indicative of mineralized bodies. Areas showing significant gravity anomalies may correlate with the presence of dense, copper-bearing sulfides, making this a key parameter for exploration.

○ **First Order Lineaments:**

- Major structural features, such as faults and shear zones, are fundamental controls on the spatial distribution of copper deposits. These first-order lineaments facilitate the movement of deep-seated mineralizing fluids and are typically associated with significant ore bodies.

XGBoost Key Features:



1. MgO (Magnesium Oxide):

- High MgO concentrations are often associated with ultramafic or mafic rocks, which can host copper mineralization. Its importance indicates that geochemical signatures in these lithologies are essential for identifying potential copper-rich zones.

2. Cr₂O₃ (Chromium Oxide):

- Chromium oxide is commonly found in ultramafic rocks and its presence can indicate environments conducive to copper deposits. This geochemical signature suggests a potential correlation between chromium-rich lithologies and copper mineralization.

3. Fe₂O₃ (Iron Oxide):

- Consistent with the Random Forest analysis, Fe₂O₃ is critical for identifying hydrothermal alteration zones. Its role in both models highlights its importance as a reliable indicator of potential copper mineralization.

4. V (Vanadium):

- Vanadium acts as a pathfinder element in specific mineralization environments, suggesting that areas with vanadium anomalies may also host copper mineralization, particularly in association with mafic and ultramafic rocks.

5. Ni (Nickel):

- Nickel's correlation with ultramafic systems indicates its potential utility as a vector for copper exploration. Areas with significant nickel concentrations could indicate environments favorable for copper deposits.

6. Zn (Zinc):

- Zinc anomalies, often found in association with base metal sulfides, may co-occur with copper mineralization. The importance of zinc in this analysis suggests it could be used to identify polymetallic environments where copper is present.

7. Second & First Order Lineaments:

- Both second-order and first-order lineaments are essential in understanding structural controls on copper localization. Their presence reinforces the significance of these features in guiding exploration efforts.

Conclusion:

The feature importance analyses from both Random Forest and XGBoost highlight several key parameters that are critical for copper prospectivity. Emphasizing **structural controls** (lineaments) and **geochemical indicators** (Fe_2O_3 , MgO , Cr_2O_3 , etc.) provides a robust framework for guiding exploration activities. By focusing on these identified features, exploration efforts can be directed towards areas with the highest potential for discovering economically viable copper deposits.

Feature Importance Study for Gold Prospectivity Using Random Forest and XGBoost

Random Forest Feature Importance

1. Th (Thorium):

- Thorium concentrations serve as a key geochemical marker, indicating the presence of geological formations that may host gold deposits. Elevated levels of thorium can highlight prospective areas for exploration.

2. Na_2O (Sodium Oxide):

- Sodium oxide is indicative of hydrothermal alteration processes. Variations in Na_2O concentrations can help identify alteration zones commonly associated with gold mineralization.

3. Copper:

- The concentration of copper acts as a direct geochemical indicator of nearby gold mineralization. High copper levels in sediments often correlate with gold presence, reinforcing its importance in exploration.

4. Lineament First Order:

- Major structural features, such as faults, control the movement of mineralizing fluids and are fundamental in locating gold deposits. These first-order lineaments enhance the understanding of spatial distribution in prospectivity.

5. Ferric Band Ratio (ASTER):

- The Ferric band ratio indicates oxidized zones that are typically associated with gold mineralization. Its significance in the analysis reflects its utility in identifying potential ore-bearing areas.

6. Geology:

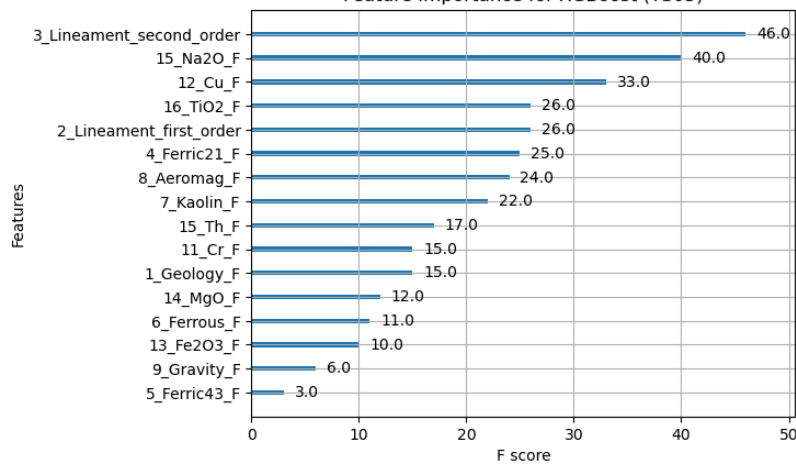
- Geological characteristics, including rock types and structures, are crucial in determining gold prospectivity. Understanding the geological context helps prioritize exploration targets.

7. RTP Total Magnetic Intensity:

- Magnetic intensity anomalies can indicate the presence of mineralized bodies. This geophysical parameter is vital for identifying potential gold deposits below the surface.

XGBoost Feature Importance

Feature Importance for XGBoost (TS05)



1. Th (Thorium):

- Similar to Random Forest, thorium plays a significant role as a geochemical indicator of prospective geological formations for gold.

2. Na₂O (Sodium Oxide):

- The importance of sodium oxide in XGBoost mirrors its role in indicating hydrothermal alteration zones, critical for gold exploration.

3. Ferric Band Ratio (ASTER):

- The Ferric band ratio provides insights into oxidized zones from satellite data, helping to identify areas where gold mineralization may be present.

4. Lineament Second Order:

- Second-order lineaments indicate minor structural features that can facilitate the movement of mineralizing fluids, emphasizing their role in gold prospectivity.

5. Copper:

- Copper concentrations continue to be a strong indicator of gold presence, reflecting its significance across both machine learning models.

6. Geology:

- Geological features remain fundamental in determining the distribution of gold deposits, highlighting the importance of geological mapping in exploration.

Conclusion

The feature importance studies using both Random Forest and XGBoost provide complementary insights into the parameters influencing gold prospectivity. Key indicators such as thorium (Th), sodium oxide (Na₂O), copper, and various lineaments consistently emerge as critical factors in both analyses. The inclusion of unique features, such as the Ferric band ratio in both models, demonstrates the value of diverse data sources and methodologies

Appendix-I

Jupyter notebooks for Random Forest and XGBoost

Below describe the build of the jupyter notebooks. Every training set (TS01, TS02..) has its own notebook for each XGBoost and Random Forest.

A total of 10 notebooks for Copper prospectivity map (CPM) - 5 for Random Forest and 5 for XGboost. Similarly for Gold prospectivity map

Imports and Setup

In the beginning, various libraries are imported:

- **Pandas** is used for data manipulation and handling data in DataFrame format.
- **Rasterio** is utilized for reading and writing raster data, which are pixel-based images.
- **Matplotlib** is for plotting and visualizing data.
- **Shapely** is used for geometric operations (like points).
- **OS** provides functionalities to interact with the operating system (like file paths).

2. Directory and File Setup

Next, a directory path is specified where all the raster files are located. A list of raster filenames is also created, naming them sequentially (from BP1.tif to BP17.tif). This sets up the framework for loading the raster data later.

3. Load Training Points

The code then loads training points from a CSV file. These points contain coordinates that will be used to extract pixel values from the rasters. Essentially, you have a set of locations where you know the classification (like land type, vegetation, etc.) based on a labeled dataset.

4. Extract Raster Values at Training Points

In this block, the code loops through each raster file, reading the raster data one by one. For each raster, it extracts the pixel values at the coordinates specified in the training points. These values are then added to a new DataFrame, which will contain all the extracted values alongside the coordinates. If a raster cannot be opened (perhaps due to a file error), it handles that situation gracefully by marking that raster's values as `None`.

5. Plotting Example

To visualize the data, the first raster is plotted, showing the raster image with red dots indicating the locations of the training points. This visual representation helps in understanding how well the training points are distributed across the raster data.

6. Saving Extracted Values

Once the raster values have been extracted and combined with the training points, the results are saved into a new CSV file. This file serves as a record of the extracted data and can be used for further analysis.

7. Restructuring Data for Random Forest Classification

Before training the machine learning model, the DataFrame is restructured:

- The columns with coordinates (X and Y) are dropped because they are not features for classification.
- The label column (the classification of each point) is moved to the end of the DataFrame. This organization is crucial because machine learning algorithms expect features (independent variables) and labels (dependent variables) to be clearly defined.

8. Random Forest Classifier and XGBoost Training

In this block:

- The feature columns (data values from rasters) and label column (classifications) are separated.
- The dataset is split into two parts: a training set (80%) and a validation set (20%). The training set is used to teach the Random Forest model, while the validation set tests how well the model performs on unseen data.
- The Random Forest classifier and XGBoost is initialized and trained using the training set. This model consists of multiple decision trees that work together to make predictions.

9. Model Evaluation

After training, the model is used to make predictions on the validation set. Various evaluation metrics are computed:

- Confusion matrices provide insight into how many predictions were correct or incorrect.
- Classification reports summarize precision, recall, and F1 scores for the model's performance.
- Accuracy and ROC AUC scores help assess how well the model distinguishes between classes.

10. Out-of-Bag Error

Since a portion of the training data is not used in building each tree in the Random Forest, an out-of-bag (OOB) score is calculated, which acts as an internal validation metric. This helps in understanding how well the model might generalize.

11. ROC Curve Plotting

The code computes the Receiver Operating Characteristic (ROC) curve, which illustrates the trade-off between true positive rates and false positive rates at various thresholds. This is useful for evaluating the model's performance across different decision boundaries.

12. Model Saving

Finally, the trained Random Forest model is saved to a file using the `joblib` library. This allows you to reuse the trained model later without needing to retrain it.

13. Prediction on New Raster Data

In this block, the code defines a function to read and stack multiple raster files into a single three-dimensional array. This allows the classifier to make predictions based on the entire stack of rasters.

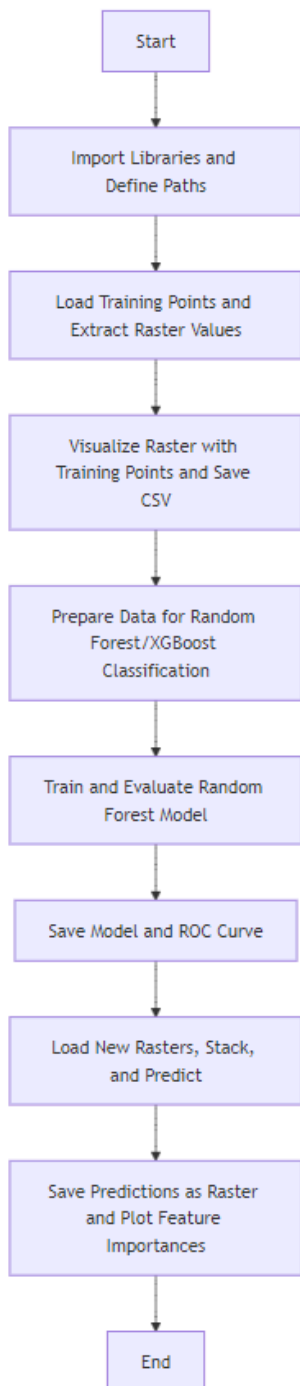
- The raster data is read, trimmed to match the smallest dimensions, and reshaped so that each pixel's values can be fed into the model.
- The predictions for each pixel are then reshaped back into a 2D array.

14. Saving Prediction as a Raster File

Finally, the predicted classifications are saved as a new raster file. This output raster can be opened in QGIS or other GIS software for visualization and further analysis.

15. Feature Importance Plotting

Lastly, the code retrieves and visualizes the importance of different features used in the model, showing which raster layers contributed most to the predictions.



Appendix-II

Folder structure for Copper prosepctivity map (CPM)

- #Readme ---> File contains the names of features for BP1 to BP17

Predictor maps / Features :

- BP1.tif
- BP2.tif
- BP3.tif
- BP4.tif
- BP5.tif
- BP6.tif
- BP7.tif
- BP8.tif
- BP9.tif
- BP10.tif
- BP11.tif
- BP12.tif
- BP13.tif
- BP14.tif
- BP15.tif
- BP16.tif
- BP17.tif

Jupyternotebooks ML_TS for Random Forest, ML_TS01_XGB for XGBoost

- CuPyML_TS01.ipynb
- CuPyML_TS01XGB.ipynb
- CuPyML_TS02.ipynb
- CuPyML_TS02XGB.ipynb
- CuPyML_TS03.ipynb
- CuPyML_TS03XGB.ipynb
- CuPyML_TS04.ipynb
- CuPyML_TS04XGB.ipynb
- CuPyML_TS05.ipynb
- CuPyML_TS05XGB.ipynb

Copper Training location sets

- CuTS01.csv
- CuTS02.csv
- CuTS03.csv
- CuTS04.csv
- CuTS05.csv

Output for Random Forest classifier as Tif

- CuTS01_prediction.tif
- CuTS02_prediction.tif
- CuTS03_prediction.tif
- CuTS04_prediction.tif
- CuTS05_prediction.tif

Output for XGBoost as Tif

- XGBTS01_cu.tif
- XGBTS02_cu.tif
- XGBTS03_cu.tif
- XGBTS04_cu.tif
- XGBTS05_cu.tif

Basic data extracted during running of code

- Cu_ML_base.csv
- extracted_raster_values.csv
- Sonpy_ML_base.csv

Trained models can reused directly as .joblib files

- random_forest_model.joblib
- xgboost_model.joblib

Appendix-III

Folder structure for Gold prospectivity map (APM)

- #Readme ---> File contains the names of features for BP1 to BP17

Predictor maps / Features:

- BP1.tif
- BP2.tif
- BP3.tif
- BP4.tif
- BP5.tif
- BP6.tif
- BP7.tif
- BP8.tif
- BP9.tif
- BP10.tif
- BP11.tif
- BP12.tif
- BP13.tif
- BP14.tif
- BP15.tif
- BP16.tif
- BP17.tif

Jupyter Notebooks for Random Forest and XGBoost

- Au_ML_TS01.ipynb
- Au_ML_TS01XGB.ipynb
- Au_ML_TS02.ipynb
- Au_ML_TS02XGB.ipynb
- Au_ML_TS03.ipynb
- Au_ML_TS03XGB.ipynb
- Au_ML_TS04.ipynb
- Au_ML_TS04XGB.ipynb
- Au_ML_TS05.ipynb
- Au_ML_TS05XGB.ipynb

Gold Training location sets

- AuTS01.csv
- AuTS02.csv
- AuTS03.csv
- AuTS04.csv
- AuTS05.csv

Output for Random Forest classifier as Tif

- AuTS01_prediction.tif
- AuTS02_prediction.tif
- AuTS03_prediction.tif
- AuTS04_prediction.tif
- AuTS05_prediction.tif

Output for XGBoost as Tif

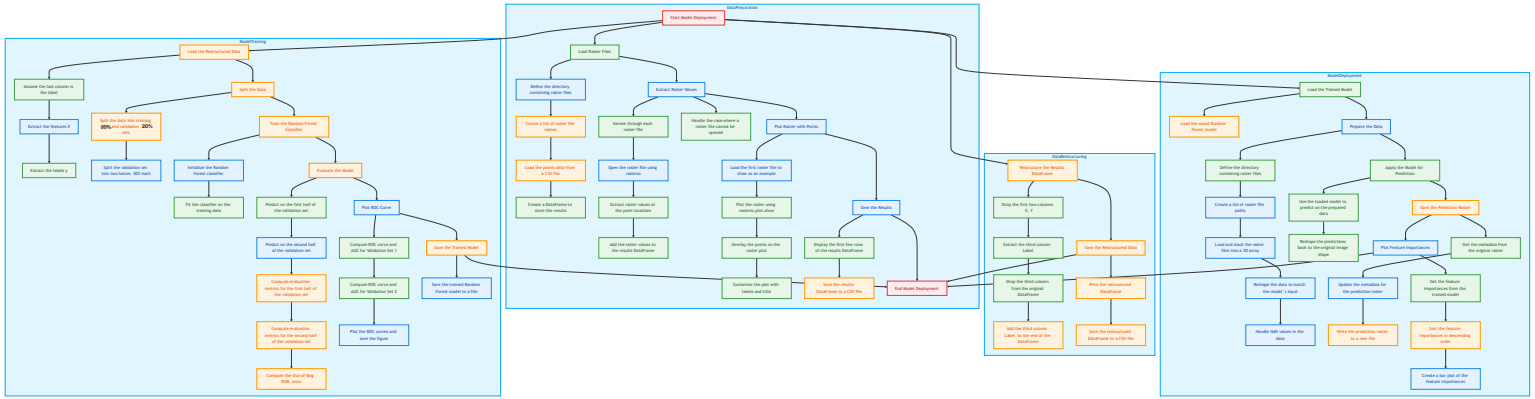
- XGBTS01_au.tif
- XGBTS02_au.tif
- XGBTS03_au.tif
- XGBTS04_au.tif
- XGBTS05_au.tif

Basic data extracted during running of code

- Au_ML_base.csv
- extracted_raster_values.csv
- Sonpy_ML_base.csv

Trained models can be reused directly as .joblib files

- random_forest_model.joblib
- xgboost_model.joblib



Recommendations and Acknowledgement

Scope for Improvement

- **Incorporate more training location data:**
 - Expanding the dataset with additional field locations would improve model accuracy and reliability.
- **Availability of geophysical maps (EM, Resistivity, Radiometric maps):**
 - Including electromagnetic (EM) and resistivity maps could enhance predictive power by providing subsurface geological information.
- **Pb-Zn prospectivity mapping:**
 - Current limitations arise from having too few training locations for Pb-Zn.
 - Data from Zawar, a known Pb-Zn region, could have been included to strengthen the prospectivity mapping.
- **Adding more machine learning ensemble models:**
 - Incorporating additional ensemble methods (e.g., Gradient Boosting, AdaBoost) can improve model performance by combining multiple learning algorithms for more robust predictions.

Acknowledgement

We would like to extend our sincere gratitude to the **Director** of the Atomic Minerals Directorate for Exploration and Research (AMD), for recommending and granting us the opportunity to participate in this hackathon. We also wish to acknowledge **Bhibu Das** for his constant guidance and expertise in processing geophysical data, which has been instrumental in our progress.

Thank you for your unwavering support.